

DC014

Computer Science 1

Semester 1

2024/2025

2 hours

DC014

Sains Komputer 1

Semester 1

Sesi 2024/2025

2 jam

Nama :												Tutoran :											
Nama Pensyarah Tutoran :																							
No. Matrik												No. Kad Pengenalan											
M	S																						



KOLEJ MATRIKULASI PERAK
PERAK MATRICULATION COLLEGE

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

SAINS KOMPUTER
2 jam

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No. Soalan	Markah	Markah Pemeriksa
1		
2		
3		
4		
5		
6		
JUMLAH		

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions.

Answer **all** questions in the space provided.

The use of non-programmable scientific calculators is permitted.

ARAHAN KEPADA CALON:

*Kertas soalan ini mengandungi **6** soalan.*

*Jawab **semua** soalan di ruang yang disediakan.*

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan

1. You are preparing a digital storytelling project for your English class. You want to incorporate multimedia elements to enhance the narrative and captivate your audience. Hence, a video is one of the multimedia elements that must be incorporated into digital storytelling.

- (a) Describe one (1) multimedia element, other than the one stated above, that you should incorporate into digital storytelling.

[2 marks]

- (b) Explain the function of the multimedia element mentioned in 1(a).

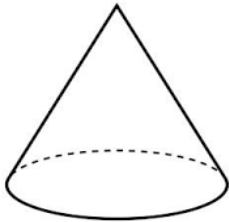
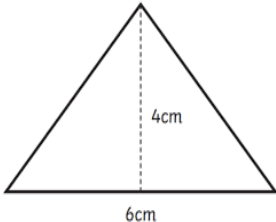
[2 marks]

- (c) Explain two (2) reasons for incorporating video in the project.

[2 marks]

2. (a) (i) Based on the shape in the table below, compare the difference between dimensions and axis in multimedia application representation.

[2 marks]

Shape		
Axis		
Dimension		

- (ii) Describe another representation of multimedia applications besides those portrayed in 2(i).

[2 marks]

- (b) A good usability of multimedia projects is constructed across three phases of development.

Examine the scenario below and identify the phases of multimedia design and the activities involved.

[6 marks]

Situation	Phase and activity
(i) The multimedia team creates visuals, buttons, and icons using the prototype screens as guidelines.	Phase: Activity:
(ii) Screen designers create the screen or interface action, which determines how the interface responds to the user.	Phase: Activity:
(iii) Ahmad creates another copy of the multimedia product from the original and packages it appropriately.	Phase: Activity:
(iv) Fatimah writes a CD-ROM or floppy disc after the multimedia development procedure is completed.	Phase: Activity:
(v) Create low-fidelity interactive prototypes with software like Adobe XD or Figma to simulate the function and user interaction with the application interface.	Phase: Activity:
(vi) One of the multimedia team members is assigned to check spelling and grammar for consistency in formatting. In addition, all videos and audio are reviewed following the scripts.	Phase: Activity:

(c) Explain the **two (2)** types of authoring tools that are shown in **Figure 1**

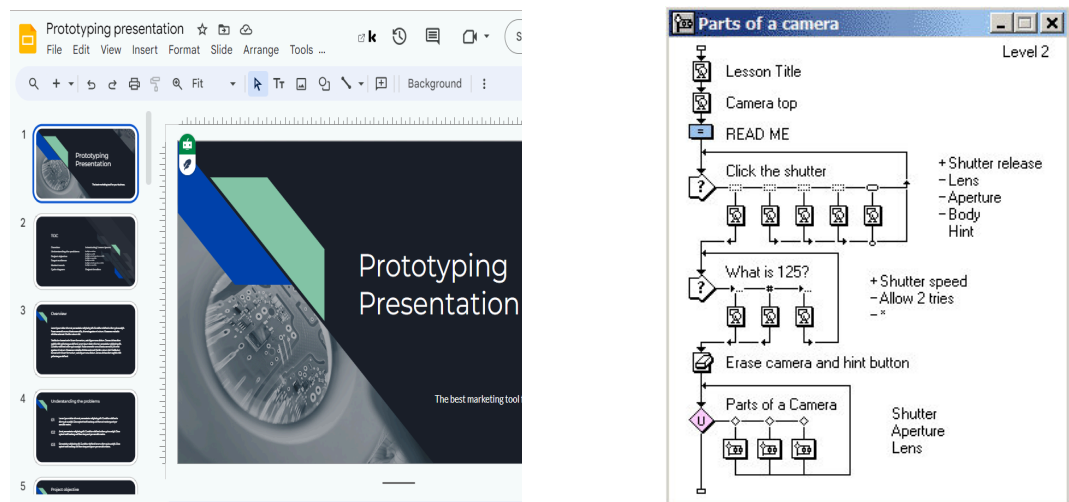
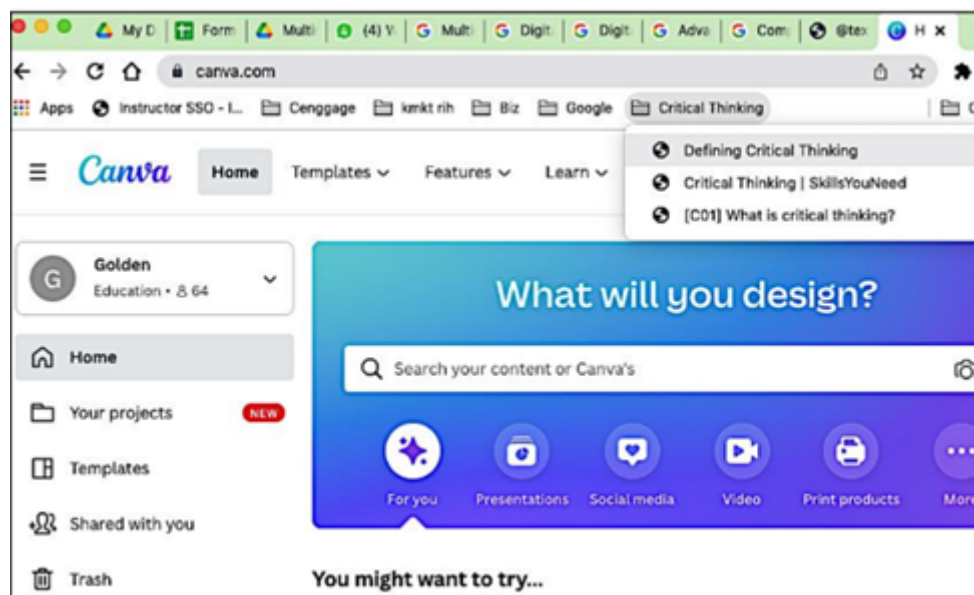


Figure 1

[4 marks]

Type of Authoring Tool	Characteristic

(d) Based on **Figure 2**, answer the question below.



Name and explain one (1) type of navigation used in **FIGURE 2**.

[2 marks]

3. (a)

Each department manages their own database
Database is design for single or limited number of users
Data collected are usually small in size/ number

Based on the aforementioned statements, identify the best approach to handling and storing information.

[1 mark]

(b) Describe **one (1)** approach that can be used to address the drawback if approach 3(a) is retained.

[2 marks]

(c) (i) Explain the **two (2)** justifications you stated for your answer in 3(b).

[2 marks]

(ii) List the **one (1)** drawback of the approach mentioned in 3(a) that necessitates switching to a better approach.

[1 mark]

4. (a)

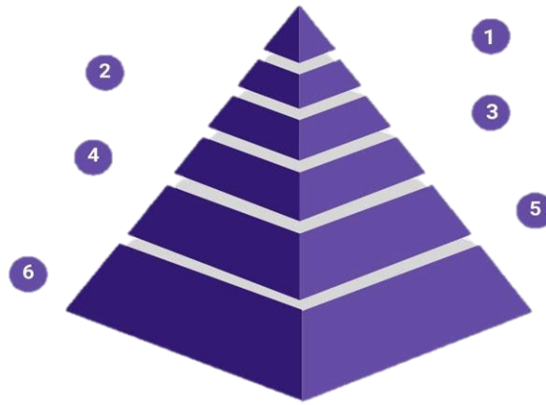


Figure 3

A data hierarchy is the structure and organization of data in a database. Data hierarchy can be visualised as a pyramid arranged based on the smallest unit to the largest that makes up a database.

Based on figure 3, write the data hierarchy name of the missing layer according to the hierarchy of data. in the table below.

[2 marks]

Layers	Data Hierarchy
1	Database
2	
3	Field
4	Character
5	
6	Bit

(b) Suggest the suitable data type to be used for the following data description.

[2 marks]

Data description	Data type
Identification card number	
Yes or No questions	

5. (a) A computer network is a collection of networked computers that communicate with one another to exchange resources and information. Computer networks are primarily used for communication, resource sharing, remote access, and collaboration.

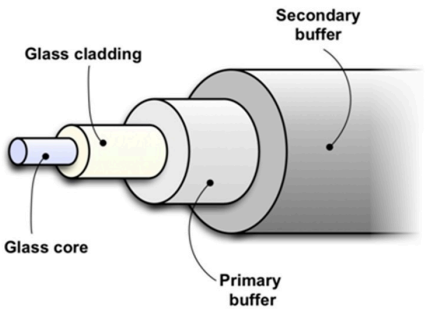
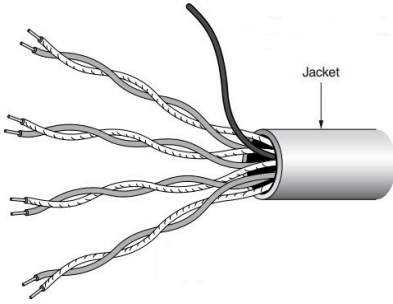
Fill in the blank cell with the appropriate area and explanation.

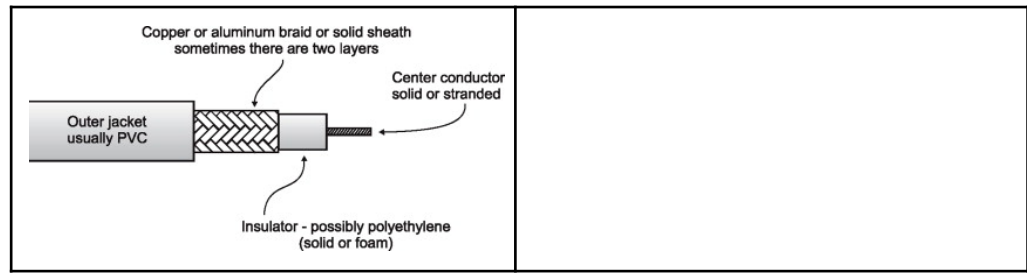
[6 marks]

Areas	Explanation
GPS network	
Application Monitoring System	
	It is a network application to deliver digital multimedia content to computers. The user usually has to have a computer or smart tv to access this service
	A voice-grade voice service employing analog signal transmission over copper wires.

- (b) State the type of wired transmission media based on the given diagram.

[3 marks]

diagram	Type of wired transmission media
	
	



(c) State the type of wireless transmission media based on the given characteristics.

[2 marks]

Diagram	Type of wireless transmission media

(d)

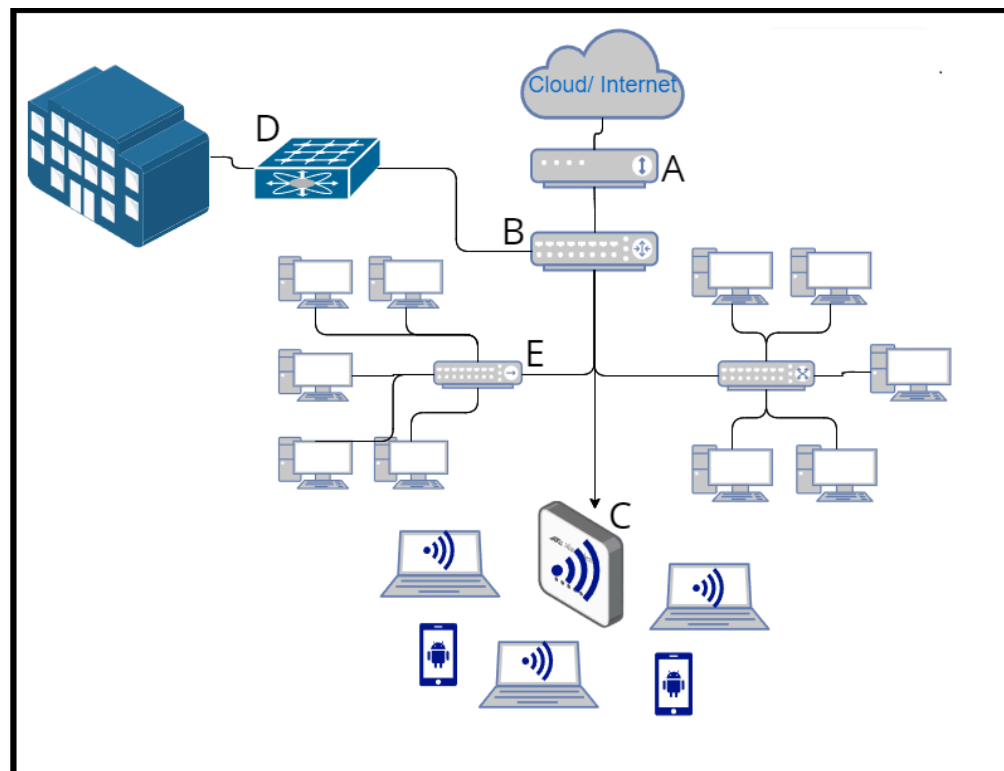


Diagram 1

Diagram 1 shows a computer network that has several of the communication devices labeled from A to E. Name the communication devices based on the label using the table below

[5 marks]

Label	Communication device
A	
B	
C	
D	
E	

(e) Name a communication device that is needed by each device so that the devices can connect to a network.

[1 marks]

6.

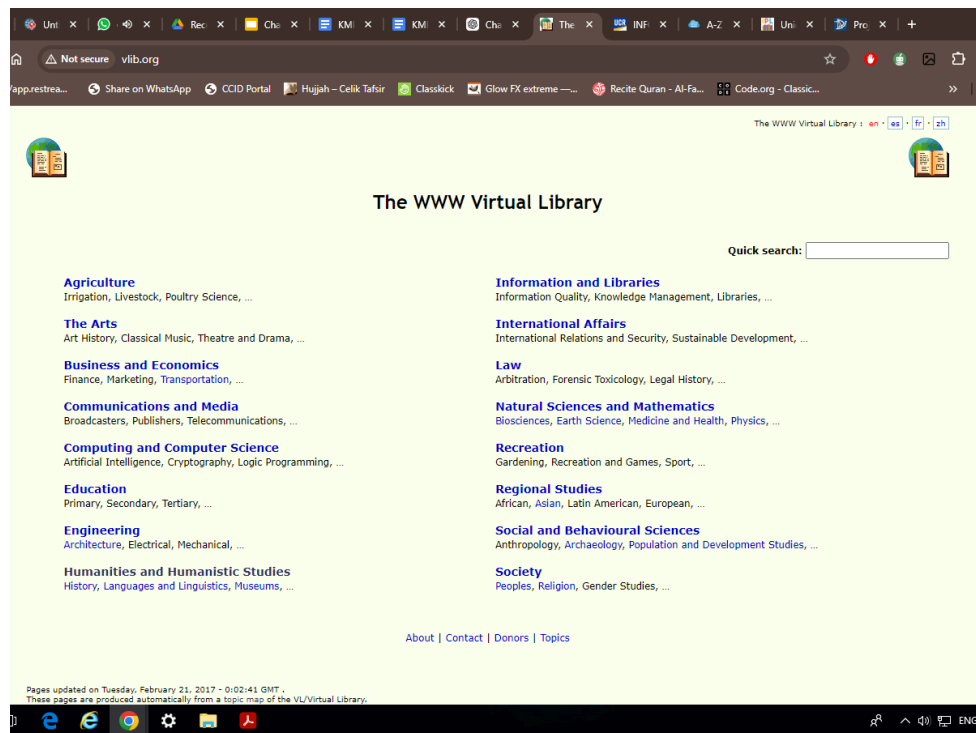


Image 1

- (a) Image 1 is the screenshot taken from a web browser showing one of the tools that can be used when searching the internet.

Name and describe the search tool.

[3 marks]

- (b) Image 2 is a screenshot of one (1) of the strategies that can be used when searching information on the internet.

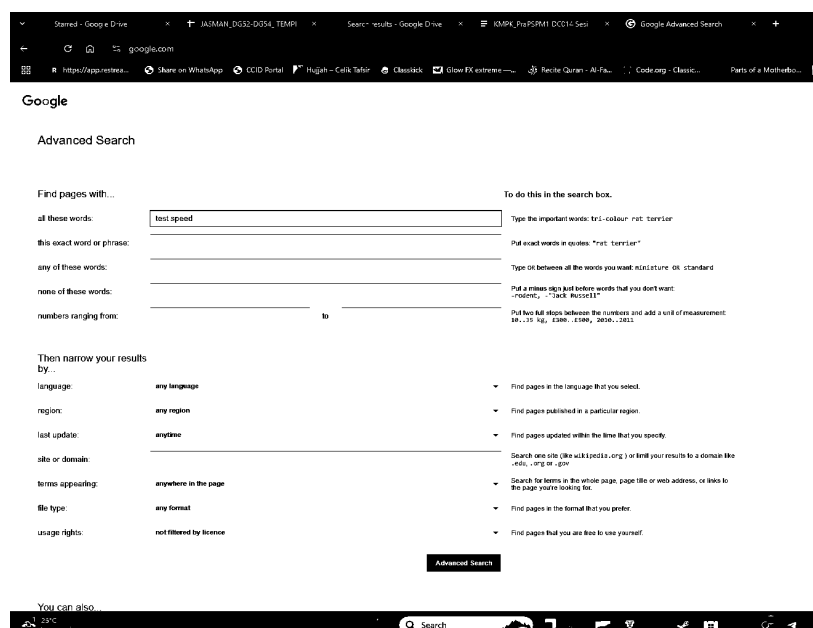


Image 2

- (i) Identify the search strategies used in image 2. Briefly describe your answer. [3 marks]

- (ii) Name three (3) search operators that can be used if you are using boolean operators in your search strategies. [3 marks]

(c)



Figure 4

Figure 4 shows a type of web activities services that provide delivery of education or any type of training by electronic methods. This electronic method can be a computer or a smartphone where teaching material is accessed by use of the internet usually.

- (i) Describe distance learning [2 marks]

- (ii) State two (2) other activities that are related with the categories of activities suggested in figure 4.

[2 marks]

- (d) (i)

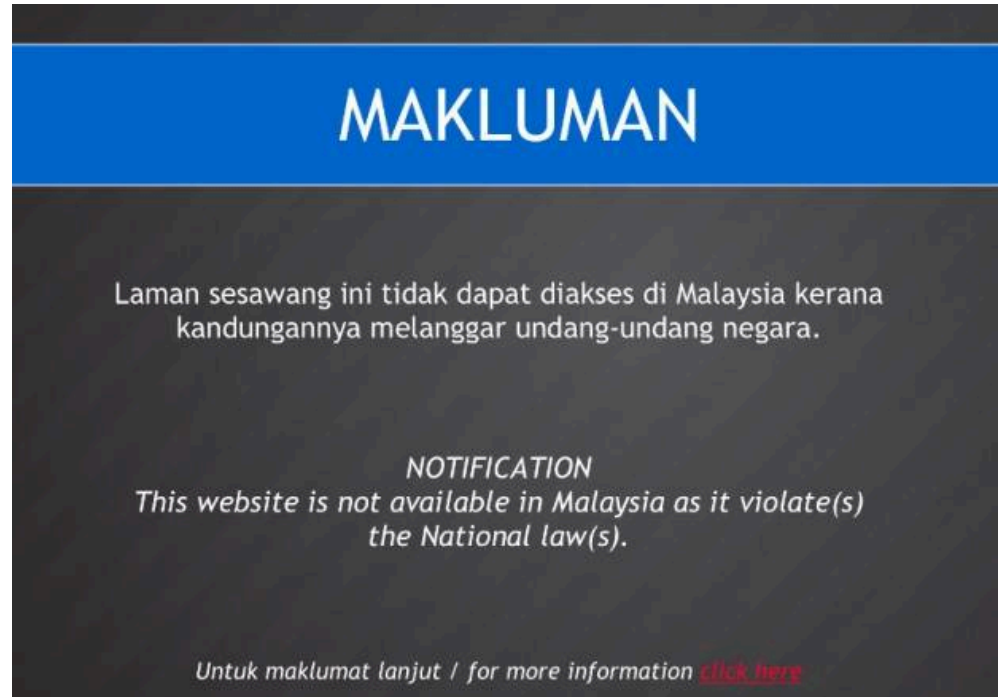


Figure 5

State the internet issue related to the image in figure 5.

[1 mark]

- (ii) Describe the answer you have stated in 6(d)(i).

[2 marks]

- (e) Data privacy is the relationship between the collection and dissemination of data, technology, the public expectation of privacy, and the legal and political issues surrounding them.

List two (2) measure user can practice to increase privacy while on the internet

[2 marks]

- (f) IoT Analytics reported there were 16.6 billion IoT devices connected in 2023. In 2020, the same firm projected there would be 30 billion devices connected by 2025. As of October, 2024, there are around 17 billion. - Wikipedia, Internet of things

Describe IoT.

[2 marks]

END OF QUESTION PAPER